The melonic ladders of $\mathcal{N} = 2$ SYK as a half-supersymmetric Dirac problem in rigid AdS_2

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Abstract

A recent construction of Thong shows that, once the centre of a conformal bilocal is integrated out, the ladder sum of a melonic CFT_d is the Robin boundary-to-boundary propagator of one free scalar in rigid AdS_{d+1}. Here we work out the analogue for $\mathcal{N} = 2$ supersymmetric SYK at the charge-symmetric point. Zero total supermomentum removes every odd relative coordinate, and the $su(1,1|1)$ pair Casimir reduces to L_0^2, which derives the eigenvalue H^2 and the shadow $H \to -H$. The reduced kernel is a single boundary correlator of a Dirac field of mass $\mu = \frac{1}{2} - 1/q$ in AdS_2, with the ladder coupling entering as a constant boundary mass, and it reproduces the known kernels with no free constant. The neutral channels require two scalars with masses $\mu(\mu \mp 1)$, which complete the Dirac field to a long AdS_2 multiplet. We show that the matrix boundary condition found in earlier work is the linear superfield condition $\Phi_+ = c\,\Phi_-$ written on the cylinder, so its boundary term is a double-trace superpotential. This answers the question of how much bulk supersymmetry survives: at $g \neq 0$ the boundary condition breaks $su(1,1|1)$, of dimension $4|4$, to $u(1|1)$, of dimension $2|2$. Beyond Thong's two broken isometries, two of the four Killing supercharges are lost, and the two that remain close on a twisted Hamiltonian. We test these statements, together with the agreement of the construction with the published kernels, spectra and four-point data, by explicit computation. We also find that the stationary bulk vacuum is supersymmetric only above a threshold $g_* < 1$, identified with a BPS bound, and that the apparent mismatch at the $H = 1$ multiplets is resolved by step-function families supported in one operator ordering.

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1 Introduction and summary

In the usual large-N dictionary, each single-trace primary corresponds to one bulk field. Melonic theories such as the SYK model [2, 3] then seem to need an infinite tower of bulk fields, one for each bilinear family. Thong has shown that, at the level of the four-point function, one field is enough [1]. Integrating out the centre of a bilocal turns the pair Casimir into an operator on a relative cylinder $\mathbb{R} \times S^{d-1}$, on which a family of dimension h is represented by the energy $\omega = h - d/2$. The ladder sum is then the Robin boundary-to-boundary propagator of a single free scalar in rigid AdS_{d+1}. The constituent dimension fixes the mass, the ladder coupling fixes the Robin coefficient, and the Robin condition breaks $2d$ isometries, leaving $\mathbb{R} \times SO(d)$.

Thong remarks that supersymmetric SYK should go beyond a single scalar [1, p. 6]. This paper studies the $\mathcal{N} = 2$ model [4], whose ladders have been solved in components [5] and in superspace [6]. It has features absent from the cases Thong treats: constituents of different dimensions, mixed neutral channels, charged channels, kernels that aren't even in h, and two distinct protected multiplets. It also raises a natural question. If the bulk field content is supersymmetric, what happens to the supersymmetry when the boundary condition breaks the AdS isometries?

Main results

1. **The cylinder (Section 3).** On a chiral–antichiral pair, zero total supermomentum leaves functions of the single invariant $x = \langle 12 \rangle$, so there is no odd relative coordinate. The $su(1,1|1)$ pair Casimir reduces to L_0^2, and $WCW^{-1} = \partial_\tau^2$. This derives the eigenvalue H^2 and shadow $-H$ of [6], and gives $\omega = H$.

	Thong, SYK ($d=1$)	$\mathcal{N}=2$ SYK (this paper)
Reduced Casimir	$\partial_\tau^2 - \frac{1}{4}$	∂_τ^2
Cylinder energy	$\omega = h - \frac{1}{2}$	$\Omega = H$
Cylinder weight	$a = 2\Delta - \frac{1}{2}$	$a = 2\Delta = \delta$
Bulk field	scalar, $m^2 = \nu^2 - \frac{1}{4}$	Dirac, $m_F = \mu$; scalars $\mu(\mu \mp 1)$
Rung	Neumann propagator	Dirac boundary correlator
Ladder sum	Robin propagator	Dirac propagator with boundary mass
Boundary condition	$B = \kappa A$	$b_F = g\sigma_p L_q\, a_F$; $\Phi_+ = c\,\Phi_-$
Spectrum	$\mathcal{R}_J(\omega) = \kappa$	$\mathcal{R}_{F,p}(\Omega) = g\sigma_p L_q$
Bulk symmetry at $g=0$	$sl(2,\mathbb{R})$	$su(1,1\|1)$
Bulk symmetry at $g\neq 0$	$\mathbb{R}$	$u(1\|1)$: $\mathbb{R}\times U(1)_R$, 2 supercharges

Table 1: Thong's construction at $d=1$ and its $\mathcal{N}=2$ analogue. Here $\nu = d - 2\Delta_\phi$ in the middle column, and $\nu = 1-\delta$, $\mu = \frac{1}{2} - \delta$, $\delta = 1/q$ in the last.

2. **A Dirac response (Section 4).** The reduced kernel is one boundary correlator of a dimension-δ fermion, with $\delta = 1/q$, multiplied by a constant boundary vertex, Eq. (4.19). Its eigenvalues reproduce the known kernels k_A, k_S with no free constant.

3. **The neutral channels (Section 5).** An ordinary centre integral keeps all components of a multiplet, which is why a 2×2 matrix with frequency-dependent eigenvectors appears. We show that its resolvent is two Dirac resolvents, Eq. (5.8), and that the matrix boundary condition is the superfield condition $\Phi_+ = c\,\Phi_-$ on the cylinder, Eq. (5.11), so the mixing matrix is derived rather than fitted.

4. **Broken supersymmetry (Section 6).** At $g \neq 0$,

$$su(1,1|1) \;\longrightarrow\; u(1|1), \qquad \text{dimension } 4|4 \to 2|2.$$

   The two broken bosonic generators are Thong's $E_\pm$. In addition, two of the four Killing supercharges are broken. The two that survive close on the twisted Hamiltonian $L_0 - \frac{1}{2}J_0$, and on them the boundary data form ordinary superfields of flat $\mathcal{N}=2$ quantum mechanics.

5. **A BPS threshold (Section 7).** The surviving algebra implies $2\omega = 2H - 1 \geq 0$ for a unitary multiplet. This is saturated at the formal coupling $g_* < 1$ where the lowest even-sector mode has zero frequency. Below g_*, the stationary vacuum breaks supersymmetry spontaneously, with vacuum energy $\frac{1}{2} - H$, whereas the four-point function on $0 < z < 1$ stays smooth.

6. **The exceptional multiplets (Section 8).** Of the two $H=1$ multiplets, one is Thong's mechanism in $\mathcal{N}=2$ form: an enhanced mode that the centre integral kills, leaving a finite remainder. The other is its mirror image. It has vanishing OPE coefficient on $0 < \chi < 1$ yet nonzero reduced weight, because the family is a finite step function supported on $\chi > 1$.

Table 1 compares the two constructions.

Scope and status

As in [1], this is a statement about four-point data at leading order in $1/N$, not an equivalence of theories. Multiparticle bulk states do not correspond to multi-trace operators, and we do not address higher-point functions or gravity. Derivations are at a physicist's level of rigour, and we don't prove convergence of the infinite family sums. Every closed-form statement below was tested numerically or symbolically, and the published formulas were transcribed and compared directly; Appendix B lists the tests. We distinguish these checked statements from conjectures, which are flagged as such. In particular, the central remaining gap is the super-centre integral of a superblock (Section 9).

Our starting point was an earlier unpublished note that proposed the matrix boundary condition and the reconstruction formula. Its formulas are confirmed here, and its central claims are given a derivation. We searched for prior work on the $\mathcal{N}=2$ extension and found none, but the search was not exhaustive. The closest precedents for the boundary-condition analysis are Amsel–Marolf [11] and Correa et al. [12]; see Section 6.6.

2 The model and its ladders

2.1 Two-point functions

The model has N complex fermions ψ_i and auxiliary complex bosons b_i, with supercharge built from an odd number $q \geq 3$ of fermions [4]. We use the component conventions of [5], with

$$\langle C_{i_1\cdots i_q} \bar{C}_{i_1\cdots i_q} \rangle = \frac{(q-1)!\, J}{N^{q-1}}. \tag{2.1}$$

In the conformal limit, at the charge-symmetric point,

$$G_\psi(\tau) = b_\psi \frac{\operatorname{sgn}\tau}{|\tau|^\delta}, \qquad G_b(\tau) = -\partial_\tau G_\psi(\tau) = \frac{\delta\, b_\psi}{|\tau|^{1+\delta}}, \qquad Jb_\psi^q = \frac{\tan(\pi\delta/2)}{2\pi}, \tag{2.2}$$

so the dimensions are $\Delta_\psi = \delta/2$ and $\Delta_b = (1+\delta)/2$. We will use throughout

$$\delta = \frac{1}{q}, \qquad \nu = 1-\delta, \qquad \mu = \tfrac{1}{2} - \delta, \qquad L_q = 2(q-1)4^{-\nu}, \qquad \sigma_p = (-1)^p. \tag{2.3}$$

Note that $0 < \mu < \frac{1}{2}$ and $\frac{1}{2} < \nu < 1$. The label $p = 0, 1$ is the parity of the fermion-pair component under exchange. The relation $G_b = -\partial G_\psi$ is the zero-supermomentum relation of Section 3 applied to the propagator.

In superspace [4, 6], the chiral field is $\Psi = \psi(\tau + \theta\bar{\theta}) + \theta b$, and correlators depend on the invariant

$$\langle 12 \rangle = \tau_{12} - 2\bar{\theta}_1\theta_2 - \theta_1\bar{\theta}_1 - \theta_2\bar{\theta}_2, \tag{2.4}$$

which is chiral in the first point and antichiral in the second. The superfield dimension satisfies $2\Delta = \delta$. The normalization of b in [6] differs from (2.2) by a factor of 2 in G_b, which accounts for 4π in place of 2π in her relation for Jb^q. Kernels are unaffected.

2.2 Ladder kernels

There are two kinds of ladder. The charged channel has ψb on one side and $\bar{\psi}\bar{b}$ on the other, with a diagonal kernel [5, (2.12)],

$$K^d(1,2;3,4) = J(q-1)\, G_\psi(\tau_{14}) G_b(\tau_{23}) G_\psi(\tau_{34})^{q-2}. \tag{2.5}$$

The neutral channels $\psi\bar\psi$ and $b\bar b$ mix, through a matrix kernel with entries [5, (2.22)–(2.24)]

$$\begin{aligned} K^{11} &= J(q-1)(q-2)\, G_\psi(\tau_{14})G_\psi(\tau_{23})G_b(\tau_{34})G_\psi(\tau_{34})^{q-3}, \\ K^{12} &= J(q-1)\, G_\psi(\tau_{14})G_\psi(\tau_{23})G_\psi(\tau_{34})^{q-2}, \\ K^{21} &= J(q-1)\, G_b(\tau_{14})G_b(\tau_{23})G_\psi(\tau_{34})^{q-2}, \qquad K^{22}=0, \end{aligned} \tag{2.6}$$

which is $K = C_0 V$ with diagonal rails C_0. We write g for a coupling multiplying each rung, with the physical theory at $g=1$. Its meaning at $g\neq 1$ is discussed in Section 7.5.

Everything is controlled by two functions [5, 6],

$$k_{A,S}(H) = -C_\delta\, \Gamma(\delta-H)\Gamma(\delta+H)\,[\sin\pi\delta \mp \sin\pi H], \qquad C_\delta = \frac{\nu\Gamma(\nu)}{\pi\delta\Gamma(\delta)}, \tag{2.7}$$

which satisfy $k_A(H)=k_S(-H)$, $k_A(n)=k_S(n)$ at integers, $k_{A,S}(1)=1$ and $k_{A,S}(0)=-(q-1)$. In terms of them, the eigenvalues of the neutral kernel at component dimension h are

$$\operatorname{spec}\mathsf{k}_1(h) = \{k_A(h),\, k_S(h-1)\}, \qquad \operatorname{spec}\mathsf{k}_0(h) = \{k_S(h),\, k_A(h-1)\}, \tag{2.8}$$

and those of the charged kernel at dimension h are $k_A(h-\frac12)$ and $k_S(h-\frac12)$. So each root of $gk_\alpha(H)=1$ gives a multiplet with component dimensions

$$(H,\ H+\tfrac12,\ H+\tfrac12,\ H+1). \tag{2.9}$$

At $q=3$, $g=1$, the first nonexceptional roots are $H_S = 2.82114\ldots$ and $H_A = 3.06590\ldots$, in agreement with the spectra of [4, 5].

We checked (2.7)–(2.8) against the printed formulas of [5] entry by entry, and against [4]. Our (2.7) has the opposite overall sign to the literal [6, (4.48), (4.49)]. Those printed signs disagree with her own (4.51) together with (B.8), with [5], and with her statement that $k=1$ at $h=1$. Appendix A lists this and the other printed discrepancies we encountered.

2.3 Two protected multiplets

At $g=1$ both branches have a root at $H=1$, with component dimensions $(1, \frac32, \frac32, 2)$. As anticipated in [4] and confirmed in [5], they differ. The S-branch multiplet gives a divergence in the conformal ladder, and is expected to correspond to the super-Schwarzian. The A-branch multiplet gives no divergence, and is expected to correspond to operators of the IR theory associated with an extra symmetry of the low-energy equations [4, Sec. VI]. We return to this in Section 8.

3 Zero supermomentum and the relative cylinder

3.1 The superconformal generators

We use the $su(1,1|1)$ generators of [6, App. A] acting on $(\tau,\theta,\bar\theta)$, for a field of dimension Δ and R-charge Q:

$$\begin{aligned} L_1 &= -\partial_\tau, \qquad L_0 = -\tau\partial_\tau - \tfrac12\theta\partial_\theta - \tfrac12\bar\theta\partial_{\bar\theta} - \Delta, \qquad J_0 = -\theta\partial_\theta + \bar\theta\partial_{\bar\theta} + Q, \\ L_{-1} &= -\tau^2\partial_\tau - \tau\theta\partial_\theta - \tau\bar\theta\partial_{\bar\theta} - 2\tau\Delta - Q\,\theta\bar\theta, \\ G_{-1/2} &= \partial_{\bar\theta} - \theta\partial_\tau, \qquad G_{+1/2} = \tau\partial_{\bar\theta} - \tau\theta\partial_\tau - (2\Delta+Q)\theta - \theta\bar\theta\partial_{\bar\theta}, \\ \bar G_{-1/2} &= \partial_\theta - \bar\theta\partial_\tau, \qquad \bar G_{+1/2} = \tau\partial_\theta - \tau\bar\theta\partial_\tau - (2\Delta-Q)\bar\theta + \theta\bar\theta\partial_\theta. \end{aligned} \tag{3.1}$$

Three remarks are needed to use these. First, we find that the printed coefficient $Q/2$ in L_{-1}, $G_{+1/2}$ and $\bar G_{+1/2}$ must read Q for the algebra to close with J_0. Second, the printed Casimir

has the translation on the left and isn't central as an operator with these generators. The central operator, with translations acting first, is

$$C = L_0^2 - \tfrac{1}{4}J_0^2 - L_{-1}L_1 + \tfrac{1}{2}(G_{+1/2}\bar{G}_{-1/2} + \bar{G}_{+1/2}G_{-1/2}). \tag{3.2}$$

Third, $Q = 0$ holds only for the pair. The two-point function $\langle 12\rangle^{-2\Delta}$ is annihilated by all total generators only for $Q_1 = -2\Delta$, $Q_2 = +2\Delta$, which is the chiral-primary condition of [4]. With these corrections, a direct implementation on the Grassmann algebra reproduces the cross-ratio form of the Casimir [6, (4.9), (A.16)],

$$\mathcal{C} = \chi^2(1-\chi)\partial_\chi^2 + \chi(1-\chi)\partial_\chi, \tag{3.3}$$

on all odd components, with $\chi = \langle 12\rangle\langle 34\rangle/(\langle 14\rangle\langle 32\rangle)$.

3.2 Reduction

Thong's first step is to restrict the pair Casimir to zero total momentum. The supersymmetric analogue is immediate.

Proposition 3.1. *Let f be a function on a chiral–antichiral pair annihilated by the total L_1, $G_{-1/2}$ and $\bar{G}_{-1/2}$. Then $f = f(x)$ with $x = \langle 12\rangle$, and*

$$C\,f(x) = L_0^2 f(x), \qquad L_0 f(x) = -(x\partial_x + 2\Delta)f(x). \tag{3.4}$$

With $x = e^\tau n$, $n = \pm 1$, and $(Wf)(\tau, n) = e^{2\Delta\tau} f(e^\tau n)$,

$$WCW^{-1} = \partial_\tau^2. \tag{3.5}$$

Proof. Chirality leaves the coordinates $(\tau_1, \bar{\theta}_1; \tau_2, \theta_2)$, which is $2|2$. The three conditions remove $1|2$, leaving $1|0$, and $\langle 12\rangle$ is the unique invariant [6]. Each term of (3.2) other than L_0^2 and J_0^2 has a translation on the right, so it annihilates f. Also $J_0 f = 0$, since $\bar{\theta}_1\theta_2$ is neutral and $Q_1 + Q_2 = 0$. Finally $\langle 12\rangle$ has weight -1, which gives the action of L_0, and W turns $-(x\partial_x + 2\Delta)$ into $-\partial_\tau$. □

So *the relative space has no odd directions*: the cylinder is the ordinary $\mathbb{R} \times S^0$, with $S^0 = \{\pm 1\}$. On $e^{H\tau}Y_p(n)$, with Y_p of parity σ_p, the eigenvalue is H^2, so $\Omega = H$ and the shadow is $-H$. This derives what [6] finds by computing (3.3) directly.

The difference from Thong is a single term. Reordering in the same way gives $L_0^2 + (1 - \mathcal{N}/2)L_0$ for $\mathcal{N} = 0, 1, 2$ at zero (super)momentum. With $L_0 \to -h$ this is

$$h(h - d_s), \qquad d_s = 1 - \frac{\mathcal{N}}{2}, \qquad \omega = h - \frac{d_s}{2}, \tag{3.6}$$

which matches the eigenvalues $h(h-1)$ [2], $h(h-\frac{1}{2})$ [9] and h^2 [6]. The number d_s is the scaling weight of the measure on the superline with $\mathcal{N}$ real odd coordinates: $d\tau\, d^{\mathcal{N}}\theta$ scales with weight $-(1 - \mathcal{N}/2)$. So $\mathcal{N} = 2$ has a dimensionless measure, which is why the Casimir has no linear term. $\mathcal{N} = 2$ is special in another way. For $\mathcal{N} = 1$, the pair coordinates are $2|2$, supertranslations remove only $1|1$, and an odd relative coordinate remains. So $\mathcal{N} = 1$ has a true super-cylinder, and it isn't simply a smaller version of this problem.

3.3 Components

Expanding (2.4),

$$f(\langle 12\rangle) = f(\tau_{12}) - 2\bar\theta_1\theta_2 f'(\tau_{12}) + \cdots , \tag{3.7}$$

so at zero supermomentum the $b\bar b$ entry of a neutral bilocal is a derivative of the $\psi\bar\psi$ entry. In the normalization of (2.2) this reads $f_b = -\partial_\tau f_\psi$. The $h = 0$ case is (2.2) itself. As a test, the matrix kernels of [5] know nothing about superspace, but act on the two-component functions $(\operatorname{sgn}\tau\,|\tau|^{h-\delta},\ c\,|\tau|^{h-\delta-1})$ and their even partners, and on the branch k_A (odd sector) or k_S (even sector) their right eigenvector has exactly

$$c = -(h-\delta) \tag{3.8}$$

for all h and q tested, which is (3.7). The other branch has

$$c = h + \delta - 1, \tag{3.9}$$

which isn't a function of $\langle 12\rangle$ alone.

This explains the structure of the neutral problem. An ordinary centre integral imposes only $L_1 = 0$, and functions annihilated by L_1 but not by $G_{-1/2}, \bar G_{-1/2}$ are still allowed. So it keeps both the bottom and the top components of each multiplet: two branches, and a matrix. Superspace keeps one function $f(x)$, one branch per parity, and one scalar kernel. We now compute that kernel.

4 The reduced ladder is a Dirac response

4.1 The reduced kernel

Take the charged ladder with seed $F_0 = G_\psi(\tau_{13})G_b(\tau_{24})$ and kernel (2.5). Write $x = \tau_{12}$, $y = \tau_{34}$, and let Y be the difference of the pair centres. As in [1], translation invariance gives fibrewise composition, $(KF)^{(0)} = K^{(0)}F^{(0)}$, on a common regulated domain.

The rail. The chain integral [1, (C.3)], with $2\Delta = \delta$, gives

$$\int dz\, G_\psi(z)G_\psi(z-t) = b_\psi^2\,\mathbf{C}\,|t|^{2\mu}, \qquad \mathbf{C} = 2B(\nu, 2\delta-1) - B(\nu,\nu). \tag{4.1}$$

The first Beta function is defined by continuation, as the integral diverges in the infrared for $q \geq 3$. Using $G_b = -\partial G_\psi$, the mixed rail is its derivative,

$$\rho(t) \equiv \int dz\, G_\psi(z)G_b(z-t) = \rho_0\,\operatorname{sgn}(t)\,|t|^{-2\delta}, \qquad \rho_0 = 2\mu\, b_\psi^2\,\mathbf{C}, \tag{4.2}$$

which defines the short-distance divergence of the boson rail by the same continuation. Notice that the odd statistics of the rail are automatic: ρ is odd, even though one of the two propagators is even.

Seed and rung. In the seed $\tau_{13} - \tau_{24} = x - y$, while in the kernel $\tau_{14} - \tau_{23} = x + y$. Since $q - 2$ is odd and $\delta(q-2) = 2\mu$,

$$F_0^{(0)}(x,y) = \rho(x-y), \qquad K^{(0)}(x,y) = \rho(x+y)\,\mathcal{V}(y), \qquad \mathcal{V}(y) = v\,\operatorname{sgn}(y)\,|y|^{-2\mu}, \tag{4.3}$$

with $v = (q-1)Jb_\psi^{q-2}$. The appearance of $x + y$ is not an accident of labelling. The vertex exchanges which rail carries the fermion, and this makes the rung act with a reflection $y \to -y$.

The same kernel follows from superspace. Acting with [6, (4.42)] on a function of $\langle 12\rangle$, set the odd coordinates of the outer points to zero. The rails then have no odd dependence, and the Grassmann integral over $\bar\theta_1, \theta_2$ acts only on the rung and f, as $\mp 2\partial_y$. Integrating by parts moves the derivative onto $\int G_\psi G_\psi \propto |x+y|^{2\mu}$, which gives (4.3). The factor of 2 is compensated by the normalization of Jb^q noted in Section 2.

The cylinder. Set $x = e^{\tau} n$, $y = e^{\tau'} n'$. Then $|x \mp y|^2 = e^{\tau+\tau'}[2\cosh(\tau-\tau') \mp 2nn']$, and the powers of $|y|$ from the measure, the rung and the weight combine as $1 - 2\mu - \delta = \delta$. This is the analogue of Thong's cancellation, and it leaves

$$v(|x||y|)^{\delta}\rho(x-y) = c\,\mathfrak{S}(\tau-\tau'; n, n'), \qquad c = v\rho_0, \tag{4.4}$$

with

$$\mathfrak{S}(t; n, n) = n\frac{\operatorname{sgn} t}{|2\sinh\frac{t}{2}|^{2\delta}}, \qquad \mathfrak{S}(t; n, -n) = \frac{n}{(2\cosh\frac{t}{2})^{2\delta}}. \tag{4.5}$$

These are the same-boundary and cross-boundary correlators of a fermionic operator of dimension δ on the two boundaries of the AdS_2 strip, which we identify in Section 4.2. The constant is

$$c = (q-1)Jb_{\psi}^{q}\, 2\mu\, \mathbf{C} = -\frac{\nu\Gamma(\nu)}{2\delta\,\Gamma(\delta)\cos(\pi\delta)\,\Gamma(1-2\delta)}. \tag{4.6}$$

On $g(\tau, n) = g_p(\tau)Y_p(n)$ the cylinder kernel preserves parity, and acts by convolution with

$$\mathbf{k}_p(t) = c\left[\frac{1}{(2\cosh\frac{t}{2})^{2\delta}} - \sigma_p\frac{\operatorname{sgn} t}{|2\sinh\frac{t}{2}|^{2\delta}}\right]. \tag{4.7}$$

Define, for $|\operatorname{Re} H| < \delta$,

$$\begin{aligned} A(H) &= \int dt\,\frac{e^{Ht}}{(2\cosh\frac{t}{2})^{2\delta}} = \frac{\Gamma(\delta-H)\Gamma(\delta+H)}{\Gamma(2\delta)}, \\ B(H) &= \int dt\,\frac{e^{Ht}\operatorname{sgn} t}{|2\sinh\frac{t}{2}|^{2\delta}} = \frac{2\cos(\pi\delta)\,\Gamma(1-2\delta)}{\pi}\Gamma(\delta-H)\Gamma(\delta+H)\sin\pi H, \end{aligned} \tag{4.8}$$

and note that $1/\Gamma(2\delta) = 2\sin(\pi\delta)\cos(\pi\delta)\Gamma(1-2\delta)/\pi$. So

$$A(H) + \sigma_p B(H) = \frac{2\cos(\pi\delta)\,\Gamma(1-2\delta)}{\pi}\,\Gamma(\delta-H)\Gamma(\delta+H)\,[\sin\pi\delta + \sigma_p\sin\pi H]. \tag{4.9}$$

Proposition 4.1. *On $e^{H\tau}Y_p(n)$, continued meromorphically from the strip $|\operatorname{Re} H| < \delta$, the reduced kernel has eigenvalue*

$$c\,[A(H) + \sigma_p B(H)] = \begin{cases} k_A(H), & p = 1, \\ k_S(H), & p = 0, \end{cases} \tag{4.10}$$

with k_A, k_S the functions (2.7). There is no free constant.

The mode $e^{H\tau}$ corresponds to the bilocal $|x|^{H-\delta}$, which is the three-point structure with the third operator at infinity, and $e^{-H\tau}$ is its shadow. Since A is even and B is odd, the shadow exchanges k_A and k_S. This is the statement $k_A(H) = k_S(-H)$, which on the cylinder is $\tau \to -\tau$. We checked (4.8) by direct integration, and (4.6), (4.10) to working precision.

4.2 The Dirac field

Consider global AdS_2, $ds^2 = \sec^2\theta\,(-dt^2 + d\theta^2)$, with $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\zeta = \cos\theta$, and a Dirac field of mass μ. With frames $\gamma^0 = i\sigma_1$, $\gamma^1 = -\sigma_3$ and $\Psi = \zeta^{1/2}\psi$, the equation $(\gamma^a\nabla_a - \mu)\Psi = 0$ becomes

$$i\partial_t\psi = H_D\psi, \qquad H_D = J_2\partial_\theta + \mu\sec\theta\,\sigma_1, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{4.11}$$

On a mode $e^{-i\Omega t}$, the rotation $\psi = e^{(\pi/4-\theta/2)J_2}(u, v)^T$ gives the first-order system

$$(\partial_\theta + \mu\tan\theta)u = (\omega+\mu)v, \qquad (-\partial_\theta + \mu\tan\theta)v = (\omega-\mu)u, \qquad \omega = \Omega - \tfrac{1}{2}. \tag{4.12}$$

At the right endpoint, $v \to a_F \zeta^{-\mu}$ and $u \to b_F \zeta^{\mu}$, so a_F has weight $\frac{1}{2} - \mu = \delta$ and b_F has weight $\frac{1}{2} + \mu = \nu$. If v has parity p, then u has parity $1 - p$. Since $|\mu| < \frac{1}{2}$ both falloffs are normalizable, and any relation between a_F and b_F compatible with the boundary form is allowed.

The regular solution of parity p fixes the response [1],

$$\mathcal{R}_{F,p}(\Omega) \equiv \frac{b_F}{a_F} = \frac{2\nu\, \mathcal{R}_{\nu,p}(\Omega - \frac{1}{2})}{\Omega - \nu}, \qquad \mathcal{R}_{\rho,p}(\omega) = \frac{\Gamma(-\rho)}{\Gamma(\rho)} \prod_{\pm} \frac{\Gamma(\frac{p+\frac{1}{2}+\rho\pm\omega}{2})}{\Gamma(\frac{p+\frac{1}{2}-\rho\pm\omega}{2})}. \tag{4.13}$$

Proposition 4.2. *The response (4.13) is the Fourier transform of the correlators (4.5):*

$$\frac{1}{\mathcal{R}_{F,p}(H)} = -\mathfrak{c}\,[\sigma_p A(H) + B(H)], \qquad \mathfrak{c} = \frac{\Gamma(\delta + \frac{1}{2})}{\sqrt{\pi}\,\Gamma(\delta)}. \tag{4.14}$$

Moreover $c = -L_q \mathfrak{c}$, so that

$$k_p(H) = \frac{\sigma_p L_q}{\mathcal{R}_{F,p}(H)}, \qquad (p, \alpha) = (1, A),\ (0, S). \tag{4.15}$$

Hence the seed is the free boundary correlator. Let $\mathfrak{G}_F = \mathfrak{c}\,\mathfrak{S}$, and let $(\mathsf{R}h)(\tau, n) = n\, h(\tau, -n)$, so that $\mathsf{R}^2 = -1$. Then (4.3) and (4.4) say

$$W K^{(0)} W^{-1} = L_q\, \mathfrak{G}_F\, \mathsf{R}, \qquad v(|x||y|)^{\delta} F_0^{(0)}(x, y) = -L_q\, \mathfrak{G}_F(\tau - \tau'; n, n'). \tag{4.16}$$

The structure is $K = C_0 V$ with the constant boundary vertex $L_q \mathsf{R}$.

4.3 The ladder

Summing the geometric series, and continuing as a resolvent as in [1],

$$v(|x||y|)^{\delta} F_g^{(0)}(x, y) = -L_q\, \mathfrak{G}_{F,g}(\tau - \tau'; n, n'), \qquad \mathfrak{G}_{F,g} = \left(\mathfrak{G}_F^{-1} - g L_q \mathsf{R}\right)^{-1}. \tag{4.17}$$

This is the $\mathcal{N} = 2$ analogue of [1, (3.11)]. No commutation of the rail and the vertex is assumed. A source j for b_F gives $b_F - g L_q \mathsf{R} a_F = j$, so the source-free boundary condition is

$$b_F = g L_q \mathsf{R}\, a_F, \qquad \text{i.e.} \qquad b_F = g \sigma_p L_q\, a_F \quad \text{on parity } p, \tag{4.18}$$

at the right endpoint. In the eigenbasis,

$$\boxed{\frac{k_p(H)}{1 - g k_p(H)} = \sigma_p L_q\, G^F_{g,p}(H), \qquad G^F_{g,p}(\Omega) = \frac{1}{\mathcal{R}_{F,p}(\Omega) - g\sigma_p L_q}.} \tag{4.19}$$

So the operator spectrum is the normal-mode spectrum, $\mathcal{R}_{F,p}(\Omega) = g\sigma_p L_q$, with $\Omega = H$.

The boundary vertex couples antipodal endpoints. In terms of endpoint data, $a_F(-) = \sigma_p a_F(+)$ and $b_F(-) = -\sigma_p b_F(+)$, so (4.18) reads $b_F(n) = n\, g L_q\, a_F(-n)$: the coupling connects the two boundaries of the strip. It is the $x + y$ of the rail that does this. See Fig. 1.

Spectral positivity. The boundary form $\bar{a}_{F,1} b_{F,2} - \bar{b}_{F,1} a_{F,2}$ vanishes on (4.18), so H_D is self-adjoint on it for real g. Thus

$$G^F_{g,p}(\Omega) = \sum_n \frac{w_n}{\Omega - \Omega_n}, \qquad w_n = \frac{1}{\mathcal{R}'_{F,p}(\Omega_n)} > 0. \tag{4.20}$$

The roots come in pairs: Ω_n at parity p and $-\Omega_n$ at parity $1 - p$, from $k_A(H) = k_S(-H)$. We confirmed $w_n > 0$ at all roots with $|\Omega| < 3.3$, for $q = 3, 5, 7$ and $g = 0.6, 0.95$.

Unlike the scalars of the next section, the Dirac field has no difficulty with stationarity at any $g \in (0, 1]$. Its boundary weight is $\delta > 0$, and its correlators (4.5) decay. The failure of Thong's condition $a + J > 0$ found for the $p = 0$ neutral sector is entirely a feature of the bosonic components (Section 7).

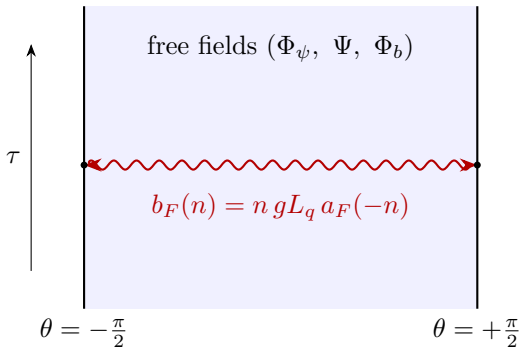


Figure 1: The auxiliary system. The bulk fields on the AdS_2 strip are free, the cylinder time τ is the logarithm of the pair separation, and the two boundaries are the two signs of the separation. All ladder rungs sit in a constant boundary vertex connecting antipodal endpoints at equal τ.

5 The neutral channels and the boundary superfields

5.1 Two scalars and a matrix condition

An ordinary centre integral of the neutral ladder (2.6) leaves a 2×2 problem. Its boundary-condition form was proposed in earlier unpublished work. We state it first, and derive its structure below. Take two scalars on the strip, solving $-\Phi'' + m^2 \sec^2\theta\, \Phi = \omega^2 \Phi$, with

$$\begin{aligned} m_\psi^2 &= \mu(\mu+1), \quad \Phi_\psi \to A_\psi \zeta^{-\mu} + B_\psi \zeta^{1+\mu}, \\ m_b^2 &= \mu(\mu-1), \quad \Phi_b \to A_b \zeta^{\mu} + B_b \zeta^{1-\mu}, \end{aligned} \tag{5.1}$$

and with parities p and $1-p$. These are $\rho^2 - \frac{1}{4}$ with $\rho = \nu$ and $\rho = \delta$. With $X = (A_\psi, B_b)^T$ and $j_0 = (-2\nu B_\psi, 2\delta A_b)^T$, regularity gives $X = \mathsf{D}_p j_0$, where

$$\mathsf{D}_p(\omega) = \mathrm{diag}\left(-\frac{1}{2\nu\, \mathcal{R}_{\nu,p}(\omega)},\ \frac{\mathcal{R}_{\delta,1-p}(\omega)}{2\delta} \right), \qquad \mathsf{M}_p = \sigma_p L_q \begin{pmatrix} 2\mu & 2\delta \\ 2\delta & 0 \end{pmatrix}. \tag{5.2}$$

Then, with $\hat{\mathsf{k}}_p$ the neutral kernel in the basis rescaled by $\mathrm{diag}(1, 1/2\delta)$,

$$\hat{\mathsf{k}}_p(h) = \mathsf{D}_p(h - \tfrac{1}{2})\mathsf{M}_p, \qquad \mathsf{G}_{g,p} = \left(\mathsf{D}_p^{-1} - g\mathsf{M}_p\right)^{-1}, \tag{5.3}$$

which we verified entry by entry against [5, (2.27)–(2.29), (2.33)–(2.35)]. The source-free boundary condition is $j_0 = g\mathsf{M}_p X$:

$$-2\nu B_\psi = c\,(2\mu A_\psi + 2\delta B_b), \qquad A_b = c\, A_\psi, \qquad c \equiv g\sigma_p L_q. \tag{5.4}$$

This condition is self-adjoint for real g, as the boundary form $2\nu(A_{\psi 1}B_{\psi 2} - B_{\psi 1}A_{\psi 2}) + 2\delta(A_{b1}B_{b2} - B_{b1}A_{b2})$ vanishes on it, even though the eigenvectors of $\hat{\mathsf{k}}_p$ depend on frequency. Using $\mathcal{R}_{\nu,p}\mathcal{R}_{\delta,1-p} = (\mu^2 - \omega^2)/(4\delta\nu)$, these eigenvectors and eigenvalues are

$$u_\pm(\omega) = \begin{pmatrix} 1 \\ (\pm\omega - \mu)/2\delta \end{pmatrix}, \qquad u_- :\ k_p(\omega + \tfrac{1}{2}), \qquad u_+ :\ k_{1-p}(\omega - \tfrac{1}{2}), \tag{5.5}$$

with $k_1 = k_A$, $k_0 = k_S$, in agreement with (2.8). Undoing the rescaling, $u_\pm$ are the eigenvectors (3.8) and (3.9).

5.2 The scalars are components of the Dirac problem

The first-order system (4.12) squares to the two scalar equations with $(\Phi_\psi, \Phi_b) = (v, u)$. The rotation with the opposite half-angle, $\psi = e^{(\pi/4+\theta/2)J_2}(u', v')^T$, gives a second such system with $\omega' = \Omega + \frac{1}{2}$, now with $(\Phi_\psi, \Phi_b) = (u', v')$ and parities $(1-p, p)$. Expanding near the right endpoint gives the boundary maps

$$\begin{aligned} \omega = \Omega - \tfrac{1}{2}: \quad & A_\psi = a_F, \quad A_b = b_F, \qquad B_b = -\frac{\omega+\mu}{2\delta} a_F, \quad B_\psi = \frac{\omega-\mu}{2\nu} b_F, \\ \omega' = \Omega + \tfrac{1}{2}: \quad & A_\psi = a_F, \quad A_b = -b_F, \quad B_b = \frac{\omega'-\mu}{2\delta} a_F, \qquad B_\psi = \frac{\omega'+\mu}{2\nu} b_F. \end{aligned} \tag{5.6}$$

Proposition 5.1. *Let ψ solve (4.11) at frequency Ω and parity p, with $b_F = c\, a_F$. Then its image at $\Omega - \frac{1}{2}$ satisfies (5.4) with c at parity p, and its image at $\Omega + \frac{1}{2}$ satisfies (5.4) with $-c$ at parity $1-p$. Their lines in the (A_ψ, B_b) plane are $u_-(\Omega - \frac{1}{2})$ and $u_+(\Omega + \frac{1}{2})$.*

Proof. For the first, $A_b = b_F = cA_\psi$, and $-2\nu B_\psi = -(\omega-\mu)ca_F = c(2\mu - \omega - \mu)a_F = c(2\mu A_\psi + 2\delta B_b)$. For the second, with $c' = -c$: $A_b = -ca_F = c'A_\psi$, and $-2\nu B_\psi = -(\omega' + \mu)ca_F = c'(2\mu + \omega' - \mu)a_F = c'(2\mu A_\psi + 2\delta B_b)$. □

So each root of $\mathcal{R}_{F,p}(\Omega) = g\sigma_p L_q$ with $\Omega = H$ gives bosonic modes at $\omega = H - \frac{1}{2}$ and $\omega = H + \frac{1}{2}$, of opposite parities. With the partner Dirac mode at $-H$ and parity $1-p$, we get the one-particle energies

$$(H - \tfrac{1}{2},\ H,\ H,\ H + \tfrac{1}{2}), \tag{5.7}$$

which is (2.9) under $\omega = h - \frac{1}{2}$. This holds for *every* g.

The same statement holds off shell. At every complex ω and real g,

$$\mathsf{G}_{g,p}(\omega)\mathsf{M}_p = \mathsf{P}_-(\omega)\, \sigma_p L_q\, G^F_{g,p}(\omega + \tfrac{1}{2}) + \mathsf{P}_+(\omega)\, \sigma_{1-p} L_q\, G^F_{g,1-p}(\omega - \tfrac{1}{2}), \tag{5.8}$$

with $\mathsf{P}_\pm$ the projectors onto $u_\pm$. So the 2×2 neutral correlator contains no information beyond the Dirac correlator. The neutral problem doesn't define an independent bulk theory; it is the Dirac response dressed with kinematic projectors. We verified (5.8) for $q = 3, 5, 7$ at real, complex and imaginary ω.

5.3 Boundary superfields

The data at an endpoint assemble into two multiplets. On the strip the boundary dimensions are $\delta - \frac{1}{2} = -\mu$ for A_ψ, δ for a_F and $1-\mu$ for B_b, and likewise μ, ν and $1+\mu$ for A_b, b_F and B_ψ. These are long $su(1,1|1)$ multiplets $(H, H+\frac{1}{2}, H+\frac{1}{2}, H+1)$ with $H = \mp\mu$, which are shadows of each other in the sense $H \to -H$. Define the top components with a Weyl shift,

$$\mathcal{B}_- = 2\delta B_b + \mu A_\psi, \qquad \mathcal{B}_+ = -2\nu B_\psi - \mu A_b, \tag{5.9}$$

and the superfields

$$\Phi_- = A_\psi + \theta\, \tilde{a}_F + \bar{\theta}\, \tilde{\bar{a}}_F + \theta\bar{\theta}\, \mathcal{B}_-, \qquad \Phi_+ = A_b + \theta\, \tilde{b}_F + \bar{\theta}\, \tilde{\bar{b}}_F + \theta\bar{\theta}\, \mathcal{B}_+, \tag{5.10}$$

where a tilde removes the factor $e^{\mp\tau/2}$ from a fermion of definite charge (Section 6.3). These are Amsel and Marolf's boundary superfields [11, (3.24)], written for $d = 2$ on the cylinder: their factors $(1 \mp 2m)$ are our 2δ and 2ν.

The shift in (5.9) is of the size expected from the superconformal map to the cylinder acting on the top component of a field of dimension $\mp\mu$, namely $-H$ times the bottom component,

although we fix it here by the computation below rather than by carrying out that map. With it, a short computation shows that (5.4) and (4.18) are together the single statement

$$\boxed{\Phi_+ = c\,\Phi_-} \tag{5.11}$$

This is the linear condition of [11] with constant $q = 1/c$. The entry 2μ of M_p, which might have looked like an accident of the reduction, is just the Weyl shift.

The boundary action is correspondingly simple. Since

$$X^T \mathsf{M}_p X = 2\sigma_p L_q\, A_\psi(\mu A_\psi + 2\delta B_b) = 2\sigma_p L_q\, A_\psi \mathcal{B}_-, \tag{5.12}$$

the quadratic term generating (5.4) is

$$-\frac{g}{2}\int d\tau\, X^T \mathsf{M}_p X = -c\int d\tau\, A_\psi \mathcal{B}_- = -\frac{c}{2}\int d\tau\, d^2\theta\, \Phi_-^2\Big|_{\text{bosons}}. \tag{5.13}$$

The double-trace term is a superpotential. The fermion bilinear in (5.13) is the one generating (4.18). At $c = 0$ and $c = \infty$ we recover the two supersymmetric single-trace conditions, $\Phi_+ = 0$ and $\Phi_- = 0$, the analogues of Thong's Neumann and Dirichlet conditions.

6 How much bulk supersymmetry survives

Thong's Robin condition breaks $2d$ AdS isometries while leaving the four-point data intact. We now ask the same question of the supercharges.

6.1 The free multiplet

Take the action on the strip,

$$S = \int d^2x\sqrt{-g}\left[\tfrac{1}{2}\Phi_b(\Box - \mu(\mu-1))\Phi_b + \tfrac{1}{2}\Phi_\psi(\Box - \mu(\mu+1))\Phi_\psi + \bar\Psi(\gamma^a\nabla_a - \mu)\Psi\right], \tag{6.1}$$

with $\bar\Psi = \Psi^\dagger\sigma_1$ and $\gamma_3 = \gamma^0\gamma^1$. The Killing equation $\nabla_a\epsilon = \frac{1}{2}\gamma_a\epsilon$ has the solutions [12]

$$\epsilon_A = \zeta^{-1/2}\begin{pmatrix}\cos(\frac{\pi}{4}+\frac{\theta}{2})\\ \cos(\frac{\pi}{4}-\frac{\theta}{2})\end{pmatrix}e^{-it/2}, \qquad \epsilon_B = \zeta^{-1/2}\begin{pmatrix}-\cos(\frac{\pi}{4}+\frac{\theta}{2})\\ \cos(\frac{\pi}{4}-\frac{\theta}{2})\end{pmatrix}e^{+it/2}, \tag{6.2}$$

giving four real supercharges with parameters ϵ and $\bar\epsilon$. For any Killing ϵ and any function Φ,

$$\begin{aligned}(\gamma^a\nabla_a - \mu)(\gamma^a\partial_a + \mu)\Phi\,\epsilon &= (\Box - \mu(\mu-1))\Phi\,\epsilon,\\ (\gamma^a\nabla_a - \mu)\,\gamma_3(\gamma^a\partial_a - \mu)\Phi\,\epsilon &= -\gamma_3(\Box - \mu(\mu+1))\Phi\,\epsilon.\end{aligned} \tag{6.3}$$

Consequently (6.1) is invariant under

$$\begin{aligned}\delta\Psi &= -\big[(\gamma^a\partial_a + \mu)\Phi_b + \beta\,\gamma_3(\gamma^a\partial_a - \mu)\Phi_\psi\big]\epsilon \;\equiv\; -M(\beta)\,\epsilon,\\ \delta\Phi_b &= \bar\Psi\epsilon + \bar\epsilon\Psi, \qquad \delta\Phi_\psi = -\beta\,\bar\Psi\gamma_3\epsilon + \bar\beta\,\bar\epsilon\gamma_3\Psi,\end{aligned} \tag{6.4}$$

for *any* constant β. These are symmetries of the free theory. Closure selects β. Acting on Φ_b,

$$[\delta_1,\delta_2]\Phi_b = \bar\epsilon_1(M - \bar M)\epsilon_2 - (1\leftrightarrow 2), \qquad M - \bar M = 2\gamma^a\partial_a\Phi_b + (\beta+\bar\beta)\gamma_3(\gamma^a\partial_a - \mu)\Phi_\psi, \tag{6.5}$$

where $\bar M = \sigma_1 M^\dagger\sigma_1$. The cross term vanishes if and only if β is imaginary, and the same normalization on Φ_ψ requires $|\beta| = 1$. So there are *two* embeddings of the superalgebra,

$$\beta = \pm i, \tag{6.6}$$

exchanged by $\Phi_\psi \to -\Phi_\psi$. In each, the anticommutators give AdS_2 Killing vectors,

$$(\epsilon_A, \bar\epsilon_A),\ (\epsilon_B, \bar\epsilon_B) \to \partial_t, \qquad (\epsilon_A, \bar\epsilon_B) \to E_+, \qquad (\epsilon_B, \bar\epsilon_A) \to E_-, \tag{6.7}$$

with $E_\pm = e^{\mp it}(i\sin\theta\,\partial_t \mp \cos\theta\,\partial_\theta)$, which take $\omega \to \omega \pm 1$ on $e^{-i\omega t}$. So the free theory has $su(1,1|1)$ symmetry, of dimension $4|4$, with bosonic generators $\partial_t, E_\pm, J_0$, where J_0 rotates Ψ. The field content $(\Phi_\psi, \Psi, \Phi_b)$ is a long multiplet with real, neutral scalars of different masses, not the chiral multiplet with a complex scalar considered in [12].

6.2 The bilinears

The maps of Proposition 5.1 are these variations. With ϵ_A, $\bar\epsilon_A\Psi$ has frequency $\Omega - \frac{1}{2}$, and with ϵ_B, $\bar\epsilon_B\Psi$ has frequency $\Omega + \frac{1}{2}$. On modes, one finds

$$(\Phi_b, \Phi_\psi) = (u, v) = (\bar\epsilon_A\psi,\ -i\,\bar\epsilon_A\gamma_3\psi), \qquad (\Phi_b, \Phi_\psi) = (v', u') = (-\bar\epsilon_B\psi,\ -i\,\bar\epsilon_B\gamma_3\psi), \tag{6.8}$$

stripped of the factors of ζ and the phases of time. The relative sign differs between A and B. In (6.4), the $\bar\epsilon$ terms have the ratio $\Phi_\psi : \Phi_b$ fixed by $\bar\beta$. Therefore:

- the first of (6.8) is the variation under $\bar\epsilon_A$ with $\beta = +i$;
- the second is the variation under $\bar\epsilon_B$ with $\beta = -i$.

In a *single* embedding, say $\beta = +i$, the $\bar\epsilon_B$ image is $(\Phi_b, \Phi_\psi) = (-v', u')$. Its line is $B_b/A_\psi = -(\omega' - \mu)/2\delta$, the opposite of the u_+ line allowed by (5.4). This is the crux of the breaking.

6.3 The surviving charges

Define, in the embedding $\beta = +i$, q_A with parameter ϵ_A and $\bar q_A$ with parameter $\bar\epsilon_A$. Let the twisted fermions be $\tilde a_F = e^{\tau/2}a_F$ and $\tilde{\bar a}_F = e^{-\tau/2}\bar a_F$ on the Euclidean cylinder, and the same for b_F. Evaluating (6.4) on modes at the boundary, we find that the charges act on the data of Section 5.3 as

$$\begin{aligned} q_A:&\quad \delta A = \tilde{\bar a}, \quad \delta\tilde a = \partial_\tau A + \mathcal{B}, \quad \delta\mathcal{B} = -\partial_\tau\tilde{\bar a}, \quad \delta\tilde{\bar a} = 0,\\ \bar q_A:&\quad \delta A = \tilde a, \quad \delta\tilde{\bar a} = \partial_\tau A - \mathcal{B}, \quad \delta\mathcal{B} = \partial_\tau\tilde a, \quad \delta\tilde a = 0, \end{aligned} \tag{6.9}$$

with the same form on $(A_\psi, \tilde a_F, \mathcal{B}_-)$ and on $(A_b, \tilde b_F, \mathcal{B}_+)$. The normalization is exactly 1 in (6.4). These are the supercharges of ordinary flat-time $\mathcal{N} = 2$ quantum mechanics, with $q_A^2 = \bar q_A^2 = 0$ and

$$\{q_A, \bar q_A\} = 2\partial_\tau \quad \text{on twisted fields,} \qquad \text{that is} \qquad 2H_{\rm tw}, \quad H_{\rm tw} = L_0 - \tfrac{1}{2}J_0. \tag{6.10}$$

A boson of frequency ω and a fermion of frequency $\Omega = \omega + \frac{1}{2}$ have the same twisted energy. In terms of the boundary superconformal generators, q_A and $\bar q_A$ are $G_{-1/2}$ and $\bar G_{+1/2}$, whose anticommutator in $su(1,1|1)$ is $2L_0 - J_0$.

Proposition 6.1. *For real $g \neq 0$, the boundary condition (5.11) is invariant under ∂_τ, J_0, and $q_A, \bar q_A$. It is not invariant under $E_\pm$ or $q_B, \bar q_B$. So, in the embedding $\beta = +i$,*

$$su(1,1|1) \ \longrightarrow\ u(1|1) = \{\, q_A,\ \bar q_A;\ L_0,\ J_0 \,\}, \qquad 4|4 \to 2|2. \tag{6.11}$$

In the embedding $\beta = -i$, the same holds with $A \leftrightarrow B$.

Generator	Action on the cylinder	$g = 0$	$g \neq 0$
L_0	∂_τ	preserved	preserved
J_0	phase of Ψ	preserved	preserved
$E_\pm$	$\omega \to \omega \pm 1$, flips parity	preserved	**broken**
$q_A, \bar{q}_A$	$\omega \to \omega \pm \frac{1}{2}$, keeps parity	preserved	preserved
$q_B, \bar{q}_B$	$\omega \to \omega \pm \frac{1}{2}$, flips parity	preserved	**broken**

Table 2: The fate of $su(1,1|1)$ in the embedding $\beta = +i$. Compared with Thong's $sl(2,\mathbb{R}) \to \mathbb{R}$, half of the supercharges survive in addition. The R-charge is preserved since (5.11) is homogeneous.

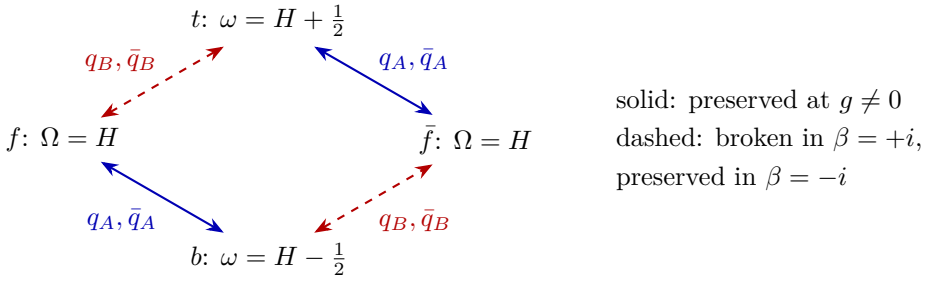


Figure 2: One multiplet in the auxiliary system at any g. The four one-particle states have energies set by a root H of $gk_\alpha(H) = 1$. Each solid edge is a two-dimensional representation of $u(1|1)$, with central charge $2H \mp 1$.

Proof. By (6.9), $q_A, \bar{q}_A$ act identically on Φ_- and Φ_+, so they preserve $\Phi_+ = c\,\Phi_-$ for constant c. The breaking of E_+ follows from the transformation of (5.4). It maps parity p to $1-p$, and $\mathsf{M}_{1-p} = -\mathsf{M}_p$, so its second component has the mismatch $-2g\omega(\mathsf{M}_p)_{21}(X_0)_1 e^{-(\omega+1)\tau}$, as found in the earlier work. The breaking of q_B follows from the line argument above, since an image with the wrong sign doesn't satisfy (5.4) unless $A_\psi = 0$ or $\omega' = \mu$. It also follows from the algebra, since if $q_B, \bar{q}_B$ together with $q_A, \bar{q}_A$ were preserved then so would be $E_\pm$, by (6.7). □

We confirmed this on explicit normal modes. The images under $(\bar{\epsilon}_A, \beta = +i)$ and $(\bar{\epsilon}_B, \beta = -i)$ satisfy (5.4) with residuals of order 10^{-19} in 20-digit arithmetic, for $q = 3, 5$ and $g = 0.7, 1$. The image under $(\bar{\epsilon}_B, \beta = +i)$ violates it by order 10. On a generic pair of scalars, the first line of (6.9) holds with normalization 1, to working precision.

Table 2 summarizes the pattern, and Fig. 2 shows the action on a multiplet.

6.4 The other charges, and the algebra they generate

The dashed maps of Fig. 2 are symmetries of the boundary condition too, but in the other embedding: q_B and $\bar{q}_B$ with $\beta = -i$ preserve (5.4) for all g. So the system with $g \neq 0$ has four odd symmetries in total, $q_A^{(+)}, \bar{q}_A^{(+)}, q_B^{(-)}, \bar{q}_B^{(-)}$. They explain why the spectrum (5.7) has the full shape of an $su(1,1|1)$ multiplet at every g, although the algebra is broken.

They aren't a broken $su(1,1|1)$ restored, however. Whereas $\{q_A^{(+)}, \bar{q}_B^{(+)}\} = E_+$, the mixed anticommutator $\{q_A^{(+)}, \bar{q}_B^{(-)}\}$ annihilates the top line, and maps the bottom line to the top line with a coefficient -4ω, in the normalization where these lines are the images of one Dirac mode under (6.8). Since this coefficient depends on the mode, the four charges do not close on a Lie superalgebra with constant structure constants and the generators listed. We have not classified what they do generate.

6.5 Origin of the breaking, and relation to the CFT

The Euclidean strip is the half-plane, $(x, z) = e^{\tau}(\sin\theta, \cos\theta)$, whose boundary is the line of relative separations x. So ∂_τ is dilatation of x, and $E_\pm$ are translation and special conformal transformation of x. This gives a simple explanation of Thong's pattern, in any d: the rung $\mathcal{V}(y) \propto |y|^{-2\nu}$ is a double-trace coupling with a profile that singles out $y = 0$ and $y = \infty$. It preserves dilatation and rotations about the origin, and breaks the $2d$ generators that move the origin. In the present case, $\mathcal{V}(y) \propto \mathrm{sgn}(y)|y|^{-2\mu}$, with a reflection, does the same.

These bulk generators aren't the CFT's pair generators, a point stressed in [1]: bulk $E_\pm$ moves between families, while the CFT's P_{12}, K_{12} move within one. The same applies to supercharges. One relation does hold. By (6.9), $\delta\tilde{a}_F = \partial_\tau A_\psi + \mathcal{B}_-$, and in terms of the original bilocal this combination is $f_b + \partial_x f_\psi$. So the condition $\delta\tilde{a}_F = 0$ is the zero-supermomentum relation (3.7), and q_A on the neutral fields is the residue of the CFT's total supertranslation $G_{-1/2}$. We have not worked out the map for the other charges. It is complicated by the reflection in the rung, as the reduced component covariance carries a factor σ_p relative to $\mathsf{G}_{g,p}$ (Eq. (8.6)).

None of this means that the CFT data lose their supersymmetry. The four-point function of Section 8 is organized by $su(1,1|1)$ superblocks at all g. It is the auxiliary bulk description that has less symmetry than the data it encodes, just as Thong's scalar has less than $SL(2,\mathbb{R})$ acting on the bulk while reproducing conformal data.

6.6 Comparison with earlier boundary-condition analyses

Amsel and Marolf [11] classify supersymmetric multi-trace conditions for scalar multiplets in AdS_d, $d = 3, 4, 5$. Our results share their main features: supersymmetric conditions relate two distinct scalars, and linear conditions generically keep half of the supercharges. They note that the scalar–fermion mass relation holds in $d = 2$ as well. There are two differences. Their preserved subalgebra is boundary Poincaré, closing on P, and their coupling is dimensionful, so it breaks dilatations. Ours closes on the twisted Hamiltonian $L_0 - \frac{1}{2}J_0$, with translations broken and dilatations preserved. In short, their condition is $\Phi_- = q\Phi_+$ in flat space with a constant, and ours is the same on the cylinder, which in flat space is a position-dependent and reflecting coupling.

Correa et al. [12] treat massless multiplets in AdS_2. Their Type I condition $\chi\alpha - \beta = 0$ also keeps 2 of 4 supercharges and breaks scale invariance. Their Type II, with a derivative in the scalar condition and an algebraic fermion condition, keeps all 4 for $m = 0$. We have not examined whether a massive analogue of Type II exists for $\Phi_\pm$, which would change the conclusions of Proposition 6.1 for a modified boundary theory, though not for the ladder data.

7 Unitarity, the threshold g_*, and the meaning of g

7.1 A BPS bound

For real g, the scalar problem is self-adjoint (Section 5), and it has no tachyons: $\omega = iy$ with real $y \neq 0$ would need $\sigma_p L_q D_\psi(\mu \pm iy)$, with D_ψ real and nonzero, to equal the real number $1/g$, which is impossible. So $\omega^2 \geq 0$ for all real g, unlike Thong's AR model, where ω^2 goes negative beyond a critical coupling [1].

The surviving algebra says more. The rotation in (4.12) is orthogonal, so $\psi^T\psi = u^2 + v^2$ pointwise, and the Klein–Gordon norm of the image of a Dirac mode is

$$2\omega \int d\theta\, (u^2 + v^2) = 2\omega\, \|\psi\|^2 = (2\Omega - 1)\, \|\psi\|^2. \tag{7.1}$$

This is the central charge of (6.10). A multiplet is unitary with the Fock vacuum as lowest state if and only if $\omega = H - \frac{1}{2} > 0$. Then every charge has the form $a^\dagger a$, and annihilates the vacuum. The bound is saturated at $H = \frac{1}{2}$.

Numerically, the only root that reaches $H = \frac{1}{2}$ for $0 < g \leq 1$ is the lowest S-branch root, which grows from $H = \delta$ at $g = 0$ to $H = 1$ at $g = 1$, as shown in Fig. 3. It does so at

$$g_* = \frac{1}{k_S(\frac{1}{2})}, \qquad g_* = 0.21784,\ 0.34729,\ 0.39485 \quad \text{for } q = 3, 5, 7. \tag{7.2}$$

This is where the lowest $p = 0$ boson has zero frequency.

7.2 Vacuum energy

Summing the zero-point energies multiplet by multiplet, with bosons $\frac{1}{2}|\omega|$ and each Dirac mode $-\frac{1}{2}|\Omega|$, gives

$$E_{\text{vac}}(H) = \frac{|H - \frac{1}{2}| + (H + \frac{1}{2})}{2} - H = \begin{cases} 0, & H \geq \frac{1}{2}, \\ \frac{1}{2} - H, & H < \frac{1}{2}. \end{cases} \tag{7.3}$$

So at $g > g_*$, including the physical point, each multiplet contributes zero, and the vacuum is supersymmetric. At $g < g_*$ the lowest multiplet contributes $\frac{1}{2} - H_S(g) > 0$. There the charges contain $a^\dagger a^\dagger$ terms, and the Fock vacuum isn't invariant: in the stationary quantization, *supersymmetry is spontaneously broken below* g_*. We sum per multiplet, and don't address regularization of the full series.

7.3 Two quantizations below g_*

At $g = 0$, Φ_ψ in the even sector has the mode $\omega = -\mu$. This is Thong's condition $a + J > 0$ failing, since $a + J = \delta - \frac{1}{2} = -\mu$. In representation theory this is the lowest state of the long multiplet with $H = -\mu < 0$. There are two options, and they differ:

- The *continued* covariance keeps the $su(1,1|1)$ covariance of the reduced seed, with a growing exponential of negative coefficient, $w(-\mu)e^{\mu|\tau|}$, $w(-\mu) = -w(\mu) < 0$, where $w(\omega) = 1/(2\nu\mathcal{R}'_{\nu,0}(\omega))$. Its norm isn't positive. This agrees with non-unitarity of the multiplet: $\|G_{-1/2}\langle H\rangle\|^2 \propto 2H < 0$.
- The *stationary* covariance has the positive decaying mode $w(\mu)e^{-\mu|\tau|}$, but no $su(1,1|1)$ invariance.

They have the same meromorphic response, and differ by a homogeneous solution,

$$G^{\text{cont}}(\tau) - G^{\text{stat}}(\tau) = -2w(\mu)\cosh(\mu\tau), \tag{7.4}$$

which we checked for $q = 3, 5$. At $g > g_*$ the question disappears, because all modes have $\omega > 0$. At the physical point, the reduced ladder computed directly from the literature's four-point function has exactly the stationary modes and weights, in the entry and range tested (Section 8.2). So no continuation in g through g_* is needed to define the physical system.

7.4 The four-point function doesn't see g_*

g_* isn't a phase transition of the correlator on $0 < z < 1$. In the superblock expansion of Section 8, poles sit at $\pm H$ and never pinch the contour, since $H \geq \delta > 0$ throughout. We confirmed that (8.3) and the root sum agree at $g = 0.15 < g_*$ for $q = 3$, as at larger g. What fails at g_* is the ordinary centre integral of a component block, which needs $\operatorname{Re} h > \frac{1}{2}$, because its tail is $|u|^{-2h}$. So g_* is a boundary of validity for the reduced bosonic description. The Dirac description, with $\Omega = H$ and weight δ, has no such boundary.

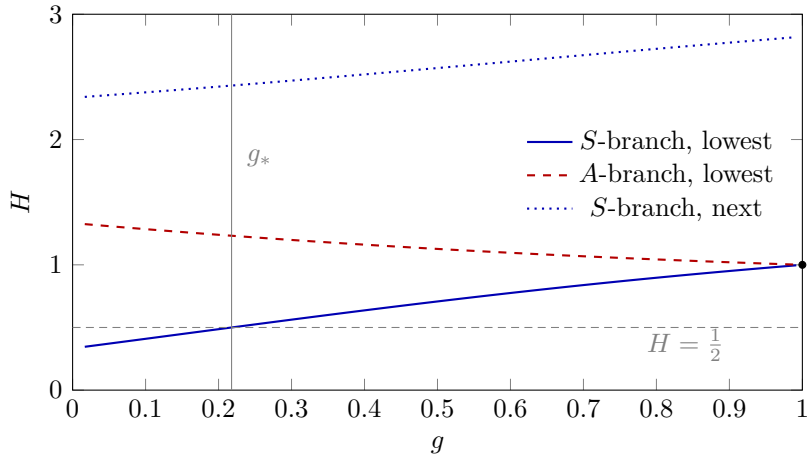


Figure 3: Lowest roots of $gk_\alpha(H) = 1$ at $q = 3$, computed from (2.7). Both lowest branches reach $H = 1$ at $g = 1$. Below g_*, the lowest S-branch root violates the bound $H \geq \frac{1}{2}$, and the stationary vacuum isn't supersymmetric. Data: `results/data/paper/spectrum_q3.dat`.

7.5 A supersymmetric line of theories

So far g has been a formal weight. In the $\mathcal{N} = 0$ model, it has a microscopic meaning through conformal SYK [8], in which the kinetic term is replaced by a bilocal. The construction appears to carry over. Replace the kinetic term by

$$S_0 = \int d\mu_1 d\mu_2 \, \bar{\Psi}(1)\, K(\langle 12\rangle)\, \Psi(2), \tag{7.5}$$

with K the inverse of $\bar{G} = \bar{b}\,\mathrm{sgn}\langle 12\rangle/|\langle 12\rangle|^\delta$, and keep the superpotential. In components we expect UV propagators $\bar{G}_\psi$ and $\bar{G}_b = -\partial\bar{G}_\psi$, though we haven't carried out this reduction. Schwinger–Dyson is then $G_i^{-1} = \bar{G}_i^{-1} - \Sigma_i$, with Σ_i as in [5]. With the conformal form, and $b_\psi/\bar{b}_\psi = b_b/\bar{b}_b = r$, we have in frequency space $G_i\bar{G}_i^{-1} = r$ and $G_i\Sigma_i = -g$, where $g = Jb_\psi^{q-1}b_b/(Jb_\psi^{q-1}b_b)_{\mathrm{IR}}$ is the same in both equations, since the IR solution makes each product -1. Thus

$$g = 1 - r, \tag{7.6}$$

which is [1, (C.9)] with $r = 2\bar{b}$. All three entries of (2.6) scale the same way, as Jb_ψ^q at fixed $b_b/b_\psi = \delta$, so $\hat{\mathsf{k}}_p \to g\hat{\mathsf{k}}_p$. This is exactly the formal g above. At $g < 1$ the $H = 1$ multiplets lift, as $k_\alpha(1) = 1 \neq 1/g$, consistent with a nonlocal theory having no local stress tensor. We have checked neither reflection positivity nor anything beyond leading order in $1/N$, so this remains a proposal.

8 The four-point function

8.1 Superconformal reconstruction

In the chiral–antichiral ordering, with cross ratio $0 < z < 1$ (the χ of [6], related to the component cross ratio of [5] by $\chi_{\mathrm{PSV}} = z/(z-1)$), the normalized seed is $F_0 = z^\delta$, and the normalized fermion covariance in this ordering is $-F$. The superblock is [6]

$$\varphi_H(z) = \frac{\Gamma(H)^2}{\Gamma(2H)} z^H {}_2F_1(H, H; 2H+1; z) = F_H(z) - F_{H+1}(z), \tag{8.1}$$

with F_h the $sl(2)$ block, $\mathcal{C}\varphi_H = H^2\varphi_H$. A fixed transform relates it to the cylinder mode, $\mathcal{T}_2 e^{-Ht} = \varphi_H$, where

$$(\mathcal{T}_2 f)(z) = 2\sqrt{2}\int_\eta^\infty \frac{\sinh(t/2)f(t)}{\sqrt{\cosh t - \cosh\eta}}\,dt, \qquad \eta = \operatorname{arcosh}(2/z-1), \tag{8.2}$$

and $\mathcal{C}\mathcal{T}_2 = \mathcal{T}_2\partial_t^2$ on this span. The four-point function is

$$F_g(z) = \int_{\mathcal{C}} \frac{dH}{2\pi i}\, r_g(H)\varphi_H(z) + \frac{2\mathcal{N}_\delta}{\pi}\sum_{n\geq 1}\frac{k_A(n)}{1-gk_A(n)}\varphi_n(z), \tag{8.3}$$

where $\mathcal{C}$ crosses the real axis in $0<c<\delta$, and

$$r_g(H) = \mathcal{N}_\delta\left[\cot\frac{\pi H}{2}\,\frac{k_A(H)}{1-gk_A(H)} - \tan\frac{\pi H}{2}\,\frac{k_S(H)}{1-gk_S(H)}\right], \qquad \mathcal{N}_\delta = \frac{\delta}{2\nu\tan(\pi\delta/2)}. \tag{8.4}$$

This follows from [6, (4.106)] after the corrections of Appendix A: send $h\to -h$ in the shadow term, and use $(1\pm\sec\pi h)/\tan\pi h = \cot\frac{\pi h}{2}, -\tan\frac{\pi h}{2}$. Closing the contour, the integer poles cancel against the explicit sum, leaving

$$F_g(z) = \sum_{k_A(H)=1/g}\frac{\mathcal{N}_\delta\cot(\pi H/2)}{g^2 k_A'(H)}\varphi_H(z) - \sum_{k_S(H)=1/g}\frac{\mathcal{N}_\delta\tan(\pi H/2)}{g^2 k_S'(H)}\varphi_H(z). \tag{8.5}$$

We verified that (8.3) equals (8.5) numerically at finite g, and that the coefficients in (8.5), together with the $F_H - F_{H+1}$ structure, reproduce the residues of [5, (4.26)] with g inserted, at $g=0.7$ and $g=1$, for $q=3,5$. That is an independent derivation of the same data in components.

8.2 The reduced component covariance

Now we can test the central claim directly, by computing the left side of Thong's identity from the literature. Let $F^{(0)}_{g,p}$ be the ordinary centre integral of the neutral correlators, projected onto unit-normalized parity harmonics, $E(\tau) = \operatorname{diag}(e^{a_\psi\tau}, e^{a_b\tau})$, with $a_\psi = \delta - \frac{1}{2}$, $a_b = \delta + \frac{1}{2}$, and $S = \operatorname{diag}(1, 1/2\delta)$. The claim, due to the earlier note, is

$$S\,E\,F^{(0)}_{g,p}\,E\,S = \sigma_p \ell_\psi\,\mathsf{G}_{g,p}, \qquad \ell_\psi = \frac{2\cdot 4^{-\nu}}{J b_\psi^{q-2}}. \tag{8.6}$$

For the $\psi\bar\psi\psi\bar\psi$ entry we can evaluate both sides independently. For the left side, we use [5, (4.26)] and Thong's reduction of a block [1, (C.8)],

$$(\mathsf{P}B_h)(\tau) \equiv \int_{-\infty}^{\infty} du\, B_h\Big(\frac{1}{u^2-\sinh^2(\tau/2)}\Big) = \mathcal{I}(h)\,e^{-(h-\frac{1}{2})|\tau|}, \qquad \mathcal{I}(h) = \frac{\pi\tan(\pi h/2)}{(2h-1)\mathcal{N}(h)}, \tag{8.7}$$

with $\mathcal{N}(h) = \Gamma(h)^2/2\Gamma(2h)$. In the even sector, which [1] doesn't need, we use Φ^a_h of [5] with the same piecewise prescription, and find by quadrature that

$$\mathcal{I}_a(h) = \frac{\pi\cot(\pi h/2)}{(2h-1)\mathcal{N}(h)}. \tag{8.8}$$

The tangents cancel against the measure, and the sum over blocks becomes

$$(\mathsf{P}F_{g,p})(\tau) = \mp 2\sum_{h_*}\operatorname*{Res}_{h=h_*}\frac{k_0^\psi(h)}{D_p(h)}\,e^{-(h_*-\frac{1}{2})|\tau|}, \qquad D_p(h) = (1-gk_p(h))(1-gk_{1-p}(h-1)), \tag{8.9}$$

with upper sign for $p=1$, and k_0^ψ the seed overlaps $k_0^{\psi,s}$ for $p=1$ and $k_0^{\psi,a}$ for $p=0$ of [5, (4.10), (4.11)]. For the right side, we use the residues of $\mathsf{G}_{g,p}$ from (5.3). The two agree at every root with $\omega<4$, for $q=3,5$, at $g=0.5, 0.9, 0.99, 1$ when $p=1$, and at $g=0.9, 0.99, 1$ when $p=0$. They include the exceptional roots below, and the full set of such tests is in Appendix B. We have not tested the entries other than the 11 entry, or the summed reduction.

Branch	p	h	Block coefficient, $g \to 1$	$\mathcal{I}$	Reduced weight
A	1	1	$\to 0$	pole	finite, nonzero
A	0	2	$\to 0$	pole	finite, nonzero
S	1	2	$\to \infty$	0	finite, nonzero
S	0	1	$\to \infty$	0	finite, nonzero

Table 3: The $H = 1$ modes. The S rows are Thong's $h = 2$ mechanism, and the A rows are its mirror image.

8.3 Exceptional families

At $g = 1$ both branches reach $H = 1$. For the S multiplet the story is Thong's for $h = 2$: the block coefficient diverges as $g \to 1$, the centre integral vanishes, and the product is finite. For the A multiplet the coefficient vanishes, since $c_A \propto \cot(\pi H/2) \to 0$ in (8.5), yet $\mathsf{G}_{1,p}$ has nonzero residues there. This looks like a contradiction with [1, (3.13)], which says that the same coefficient multiplies each family before and after reduction. It isn't, as we now show.

Φ^s_h of [5] has poles at odd h, and Φ^a_h at even h, *on* $\chi > 1$ *only*. With $x = (\chi - 2)/\chi$,

$$\operatorname*{Res}_{h=2n+1} \Phi^s_h = -2P_{2n}(x), \qquad \operatorname*{Res}_{h=2n+2} \Phi^a_h = +2P_{2n+1}(x), \qquad \chi > 1, \tag{8.10}$$

which we checked for the first two values of each, with P_n the Legendre polynomials. The poles are cancelled by zeros of the measure, as noted for Ψ_h in [2]. So on $0 < \chi < 1$ the family is absent, but on $\chi > 1$ a finite contribution remains:

$$\frac{2h-1}{\pi \tan(\pi h/2)} \Phi^s_h \xrightarrow{h\to 1} \theta(\chi > 1), \qquad \frac{2h-1}{\pi \tan(\pi(h-1)/2)} \Phi^a_h \xrightarrow{h\to 2} -3P_1(x)\,\theta(\chi > 1). \tag{8.11}$$

Lemma 8.1. *With* $\chi = 1/(u^2 - \sinh^2(\tau/2))$,

$$\mathsf{P}\big[\theta(\chi > 1) P_n(x)\big] = \frac{2}{2n+1} e^{-(n+\frac{1}{2})|\tau|}. \tag{8.12}$$

Proof. The region $\chi > 1$ is $\sinh \frac{|\tau|}{2} < |u| < \cosh \frac{\tau}{2}$, on which $x = 1 - 2(u^2 - \sinh^2 \frac{\tau}{2})$ runs over $[-1, 1]$. Changing variables gives $\int_{-1}^{1} P_n(x)\, dx\, (2\cosh\tau - 2x)^{-1/2}$. With $t = e^{-|\tau|}$ we have $2\cosh\tau - 2x = (1 - 2xt + t^2)/t$, so the Legendre generating function gives $(2\cosh\tau - 2x)^{-1/2} = \sqrt{t} \sum_m t^m P_m(x)$, and orthogonality finishes the proof. □

For $n = 0$ this is the statement that the interval has width $2(\cosh \frac{\tau}{2} - \sinh \frac{|\tau|}{2}) = 2e^{-|\tau|/2}$, so the step function reduces exactly to the mode at $\omega = \frac{1}{2}$. Together with (8.11) this reproduces (8.9) at both A-modes. Table 3 summarizes.

So the four-point function does contain the A-modes, as a step function that vanishes on $0 < \chi < 1$. The chiral–antichiral reconstruction lives at $\chi_{\text{PSV}} < 0$, which can't see it. The two statements are compatible, and [1, (3.13)] holds, provided the family at $g = 1$ is understood as this step function or as the limit $g \to 1^-$. This fits the expectation of [4] that the A multiplet has no divergence and corresponds to operators of the IR theory, and it supplies its form in the four-point function. We haven't examined what a contribution in only one ordering means physically.

8.4 The S multiplet

The divergence as $g \to 1$ reflects the soft modes. A definite conformal subtraction is [6]

$$F_{\text{sub}}(z) = \lim_{g\to 1^-} \left[F_g(z) - \frac{D_\delta}{1-g} \varphi_1(z) \right], \qquad D_\delta = \frac{2\mathcal{N}_\delta}{\pi}, \tag{8.13}$$

and with $b_1 = k_S'(1)$, $b_2 = k_S''(1)$ the pole moves as $H(g) = 1 + (1-g)/b_1 + \cdots$, leaving the finite contribution

$$\frac{D_\delta}{b_1}\partial_H \varphi_H(z)\Big|_{H=1} + D_\delta\Big(1 - \frac{b_2}{2b_1^2}\Big)\varphi_1(z), \tag{8.14}$$

whose coefficients we checked. This is a conformal finite part, not the finite-coupling answer. A different subtraction changes the coefficient of φ_1. Its physical value requires the super-Schwarzian [4], which has not been computed in any work we have read. On the reduced side the ambiguity disappears, in the way of [1, (C.12)]: $\varphi_1 = F_1 - F_2$, and $\mathcal{I}_a(1) = \mathcal{I}(2) = 0$. Accordingly the residues of $\mathsf{G}_{1,p}$ at these modes are finite and unambiguous.

9 Discussion

We have shown that the ladder structure of $\mathcal{N} = 2$ SYK reduces to a single Dirac response on the AdS_2 strip, with two scalars as its components in the bosonic reduction, and that a constant double-trace superpotential on the cylinder breaks the bulk superalgebra from $su(1,1|1)$ to $u(1|1)$. The spectrum retains the shape of the full multiplets at all g, as in Thong's case where the dimensions and OPE data survive the loss of $E_\pm$. We list what we consider the main open points.

The super-centre integral. The main remaining gap is to compute the integral of a superblock over the super-centre, $(Y, \theta_Y, \bar\theta_Y)$. The derivation in Section 4 gives the reduced seed and kernel, but the family-by-family reduction has been done only for ordinary centre integrals of component blocks. We expect one exponential $e^{-H|\tau|}$ per parity, and that $\mathcal{T}_2$ of (8.2) is the inverse map. Establishing this would make precise the statement that one Dirac field is the $\mathcal{N} = 2$ replacement of Thong's single scalar. For the moment that statement is supported by (4.19) and (5.8), but it's a conjecture at the level of the full four-point function.

Term-by-term reduction. We have not proved the analogue of [1, App. C.2] for the $\mathcal{N} = 2$ families, including the step-function families, or that the mixed ψ–b continuation allows the required exchanges of integrals. Working with $G(\langle 12\rangle)$ directly may avoid the separate infrared and ultraviolet divergences of the fermion and boson rails.

A modified boundary theory. Proposition 6.1 concerns the given boundary domain. A massive analogue of the Type II condition of [12] might preserve more, at the cost of a derivative term in the scalar condition. It isn't clear that such a theory would reproduce the ladder data.

The physical $H = 1$ coefficient. Connecting (8.14) to the super-Schwarzian [4, 13] is a well-defined next computation, which the literature we have read leaves undone [5, 6].

Antipodal coupling. The boundary vertex couples the two boundaries of global AdS_2 at equal time (Fig. 1). That has the form of the double-trace couplings that make wormholes traversable [15, 16]. We mention this only as a structural similarity. Here the bulk is rigid and the fields are auxiliary, and the two boundaries are the two signs of a relative separation, not two copies of a system.

Stepping stones. Complex SYK at zero charge is a useful test. We find its symmetric-channel kernel [7, 10] satisfies

$$k^S(h)\,\mathcal{R}_{\nu,0}(h - \tfrac{1}{2}) = 4^{-\nu}, \qquad \nu = 1 - \frac{2}{q}, \tag{9.1}$$

to be compared with $k^A\mathcal{R}_{\nu,1} = (q-1)4^{-\nu}$ [1]. So it is one scalar with a parity-dependent Robin coefficient, with a mode at $h = 1$ for the $U(1)$ charge and $a + J = 2/q - \frac{1}{2} < 0$ for $q > 4$. It is the simplest system where the even-sector issues of Section 7 arise without mixing. $\mathcal{N} = 1$ tests something else, because it has an odd relative coordinate.

Beyond this paper. Nonzero charge [10, 14] would need a θ-dependent mixing. Higher-point data remain out of reach. Thong's caveat stands: this construction concerns four-point data, not an equivalence of theories.

A Labels and printed discrepancies

Labels. The conventions in the literature collide, as follows.

This paper	[5], kernels	[5], four-point	[4]
$p = 1$ (odd fermion pair)	$k^{a,\pm}$	f^s, Φ^s	
$p = 0$ (even fermion pair)	$k^{s,\pm}$	f^a, Φ^a	
$k_A(H)$	$k^{a,+}(H) = k^{a,d}(H+\frac{1}{2}) = k^{s,-}(H+1)$		$k_a(H+\frac{1}{2})$
$k_S(H)$	$k^{s,+}(H) = k^{s,d}(H+\frac{1}{2}) = k^{a,-}(H+1)$		$k_s(H+\frac{1}{2})$

Also, the χ of [6] is our z, while the χ of [5] is $z/(z-1)$, and $\delta = 2\Delta$ in the notation of [6]. The coupling J is normalized differently in [5] and [4].

Printed equations in [6]. We list the following as algebraic inconsistencies that can be demonstrated from the paper. They are not an erratum, and none affects her structural conclusions. All were checked by computation.

Equation	Observation
(4.48), (4.49)	Overall sign. It disagrees with (4.51) with (B.8), with [5], and with $k(1) = 1$.
(4.39)	The term in φ_{-h} should have a minus sign, as follows from (4.38), (B.4) and (B.5). We confirmed (B.4), (B.5) by direct integration of (4.26).
(4.90)–(4.92), (4.106)	The prefactor should be $\alpha/2$, as (4.85) with (4.88) implies and as the seed requires; (4.99) is correct as printed.
(A.3), (A.5), (A.7)	$Q/2$ should be Q for the algebra to close with (A.4).
(A.9)	The ordering should be $-L_{-1}L_1$ for the operator to be central.
(A.15)	$Q = 0$ holds only for the pair: $Q_1 = -2\Delta$, $Q_2 = +2\Delta$.

The first two of these were pointed out in the earlier note. As far as we know, they haven't appeared elsewhere, but we have not made a thorough search.

B Record of checks

All numerical statements can be reproduced from the scripts below, in `scripts/` of the project repository, using `mpmath` and `sympy`. Each exits with an error if a check fails. Logs are in `results/data/`. Passing means agreement within the stated tolerance at the tested parameter values, generally $q = 3, 5$ or $q = 3, 5, 7$. It is not a proof.

Script	Tests
`check_literature_inputs.py`	Formulas transcribed from [5, 6, 4, 2] against Section 2; printed spectra; (3.8), (3.9); (8.6) via (8.9); (8.5) against [5]; (8.8), (8.10), (8.12); complex SYK.
`check_supercasimir.py`	The generators and (3.2): centrality, closure, invariance of the two-point function, (3.3) on all odd components, Proposition 3.1.
`check_dirac_reduction.py`	(4.8) by quadrature; (4.6); Propositions 4.1 and 4.2; positivity in (4.20).
`check_bulk_susy_algebra.py`	Symbolic: (6.2), (6.3), closure and β, (6.7).
`check_bulk_susy_modes.py`	On normal modes: (6.8), the lines of Proposition 5.1, boundary conditions at finite g for all embeddings, (6.9) in part, kernels, $\{q, \bar{q}\}$.
`check_ward_and_vacuum.py`	(5.8); the quadratic form in (5.13); (7.2), (7.3); data for Fig. 3.
`check_reconstruction.py`	(8.3) against (8.5) at finite g, including below g_*.
`check_robin_identities.py`	Thong's closed forms; (8.2); (7.4); (8.14); order g^1 of (8.3) against a direct integral.

Not checked. A direct two-rung integral at order g^2. The entries of (8.6) other than the 11 entry. The second line of (6.9), and the $\delta\mathcal{B}$ variations for generic data rather than on modes. Convergence of infinite sums. The fermion bilinear in (5.13), whose form we inferred from (4.18). The super-centre integral.

Derived by hand only. The reduction from [6, (4.42)] in Section 4, (3.6) at $\mathcal{N} = 1$, the normalization on Φ_ψ in the closure of (6.4), and Section 7.5.

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